STAT 544: Conditional Distributions II Study Guide and Lecture Prep

This guide develops conditional distributions from probability fundamentals through calculation, interpretation, and the measure-theoretic framework. The central progression is **joint law → marginal law → conditional law → conditional moments → unconditional checks**. Read the review first, work through the six problems without skipping the support calculations, and use the lecture-prep checklist to identify what still needs clarification.

The notation $E[Y X = x]$ is interpreted here as $E[Y \mid X = x]$, "the expected value of Y, given that $X = x$." This is a preparation document, not an instructor-issued syllabus: the advanced material is labeled so it can be matched to the actual lecture. The earlier Part I guide was used for continuity; its linked GitHub PDF could not be retrieved, so the prior conversation's Markdown copy was consulted instead.

Prerequisite review: the language behind conditioning

Outcomes, events, and probability

A probability model begins with $(\Omega, \mathcal{F}, P)$: a set of possible outcomes, a collection of measurable events, and a probability measure assigning those events numbers in $[0, 1]$. The measure-theoretic treatment of conditional expectation uses exactly this setup, with a smaller sigma-algebra representing the information being conditioned on (CMU probability notes).

- **Sample space Ω:** The set of elementary outcomes ω.
- **Sigma-algebra $\mathcal{F}$:** A collection containing Ω, closed under complements and countable unions. These are the events to which the model assigns probabilities.
- **Probability measure P:** A nonnegative, countably additive set function with $P(\Omega) = 1$.
- **Random variable X:** A measurable function $X : \Omega \to \mathbb{R}$. Measurability ensures that $\{X \in A\}$ is an event whenever A is a Borel set.
- **Borel sets $\mathcal{B}(\mathbb{R})$:** The sigma-algebra generated by open intervals. It includes intervals, individual points, and the usual sets used in probability calculations.
- **Indicator $\mathbf{1}_A$:** A random variable equal to 1 on event A and 0 otherwise, with $E[\mathbf{1}_A] = P(A)$.
- **Almost surely, or a.s.:** Equality or a property holds except on a set of probability zero.

These definitions let a statement about numerical values, such as $X \leq 2$, become a statement about outcomes, $\{\omega : X(\omega) \leq 2\}$. They also distinguish a random variable from its realized value: X is the map, while x is a possible number.

Distributions, mass, density, and cumulative probability

The distribution, or law, of X is the measure $P_X(A) = P(X \in A)$; the joint law is $P_{X,Y}(C) = P((X,Y) \in C)$. The marginal law is the joint law with the other coordinate unrestricted, for example $P_X(A) = P_{X,Y}(A \times \mathbb{R})$ (CMU probability notes).

For a discrete X with finite or countable value set S_X, the probability mass function is

$$p_X(x) = P(X = x), \qquad p_X(x) \geq 0, \qquad \sum_{x \in S_X} p_X(x) = 1.$$

An absolutely continuous X has a probability density function satisfying

$$f_X(x) \geq 0, \qquad \int_{\mathbb{R}} f_X(x)\,dx = 1, \qquad P(X \in A) = \int_A f_X(x)\,dx.$$

The PMF gives a probability at a point; the PDF gives probability through integration, not through its height at a point (Penn State probability distributions). In particular, $P(X = x) = 0$ for an absolutely continuous variable, and a PDF is allowed to exceed 1.

For a discrete pair, the corresponding joint definition is

$$p_{X,Y}(x,y) = P(X = x, Y = y), \qquad p_{X,Y}(x,y) \geq 0, \qquad \sum_x \sum_y p_{X,Y}(x,y) = 1.$$

For an absolutely continuous pair, the joint PDF satisfies

$$f_{X,Y}(x,y) \geq 0, \qquad \iint_{\mathbb{R}^2} f_{X,Y}(x,y)\,dx\,dy = 1,$$

$$P((X,Y) \in C) = \iint_C f_{X,Y}(x,y)\,dx\,dy.$$

These are the two-dimensional counterparts of the univariate definitions. In particular, a joint PDF is not the point probability $P(X = x, Y = y)$.

The CDF works for discrete, absolutely continuous, mixed, and singular laws:

$$F_X(t) = P(X \leq t).$$

For a discrete law,

$$F_X(t) = sum_xletp_X(x);$$

for an absolutely continuous law,

$$F_X(t) = int_-infty^t f_X(x), dx.$$

The word “continuous” needs care. Having a continuous CDF does not, by itself, guarantee a PDF; throughout the density calculations below, “continuous case” means that the required law is absolutely continuous with respect to Lebesgue measure.

Support and integration bounds

For calculation, describe the region S where the joint mass or density is nonzero, and carry its indicator as part of the formula:

$$f_{X,Y}(x,y) = g(x,y)\mathbf{1}_S(x,y).$$

The formula $g(x,y)$ alone is not the distribution. For continuous laws, changing density values on a Lebesgue-null set does not change the law, so boundary conventions such as $<$ versus $\leq$ usually do not affect integrals.

Define the two types of slices:

$$S_x = \{y : (x,y) \in S\}, \qquad S^y = \{x : (x,y) \in S\}.$$

- **Condition on $X = x$:** Hold x fixed and let y range over S_x.
- **Condition on $Y = y$:** Hold y fixed and let x range over S^y.
- **Marginalize to X:** Add or integrate over y, not over x.
- **Marginalize to Y:** Add or integrate over x, not over y.

As a geometric rehearsal, take $S = \{(x,y) : 0 < x < y < 1\}$. A vertical slice has $x < y < 1$, while a horizontal slice has $0 < x < y$; the bounds change when the integration order changes.

Expectations, variance, and the calculus you need

Expectation is a probability-weighted average, and it can be calculated by summing or integrating values against their distribution (Penn State probability distributions):

$$E[g(Y)] = \sum_y g(y)p_Y(y) \quad \text{or} \quad \int_{\mathbb{R}} g(y)f_Y(y)\,dy.$$

For a finite signed expectation, a safe assumption is $E[|g(Y)|] < \infty$. For variance, assume $E[Y^2] < \infty$, and use

$$\operatorname{Var}(Y) = E[(Y - E[Y])^2] = E[Y^2] - (E[Y])^2.$$

For variables with finite second moments, define covariance and, when both variances are positive, correlation:

$$\operatorname{Cov}(X,Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y],$$

$$\rho = \operatorname{Corr}(X,Y) = \frac{\operatorname{Cov}(X,Y)}{\sigma_X\sigma_Y}, \qquad \sigma_X = \sqrt{\operatorname{Var}(X)}, \quad \sigma_Y = \sqrt{\operatorname{Var}(Y)}.$$

Independence implies zero covariance when these moments exist. The converse is not generally true, as the last worked problem will demonstrate.

The calculations in this guide use a small toolkit:

- **Power rule:** $\int_a^b t^k\,dt = (b^{k+1} - a^{k+1})/(k+1)$, when applicable.

- **Dummy variables:** In $\int y f(x, y)\, dy$, x is fixed; the integration removes y.
- **Piecewise integrals:** Split a domain where its boundary formula changes.
- **Completing the square:** $v^2 - 2av = (v - a)^2 - a^2$.
- **Tonelli's theorem:** Nonnegative integrands permit exchanging sums or integrals, even before finiteness is established.
- **Fubini's theorem:** Absolute integrability permits exchanging integration order for signed integrands.

For example, the integral defining f_X is nonnegative, so Tonelli is the natural justification. A calculation involving $y f_{X,Y}(x, y)$, which may change sign, is justified by $E|Y| < \infty$.

Ordinary conditional probability and independence

When $P(B) > 0$, ordinary event conditioning is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

It restricts attention to B, then rescales its probability to 1; the discrete conditional-PMF formula follows by taking $A = \{Y = y\}$ and $B = \{X = x\}$ (MIT conditional expectation lecture). For a continuous X, this event formula cannot be applied directly to $\{X = x\}$, because its denominator is zero.

Independence means

$$P(X \in A, Y \in B) = P_X(A) P_Y(B) \quad \text{for all Borel } A, B.$$

In the discrete case this is equivalent to $p_{X,Y}(x, y) = p_X(x) p_Y(y)$; with a joint PDF it is equivalent to $f_{X,Y}(x, y) = f_X(x) f_Y(y)$ almost everywhere (CMU probability notes). Factorization must include the support indicator, not just the algebraic expression.

Do not replace the factorization test with "the support looks rectangular." A rectangle does not establish independence, and a restriction described loosely as "triangular" needs examination of where the density is actually positive; positive density throughout a nondegenerate triangle does rule out independence because the product of its positive marginals would put mass outside the triangle.

Standard conditional distributions

Discrete PMFs: from joint to marginal to conditional

For a joint PMF $p_{X,Y}$, first calculate the marginal of the conditioning variable:

$$p_X(x) = \sum_{y \in S_x} p_{X,Y}(x, y).$$

Then, for $p_X(x) > 0$, the conditional PMF is (MIT conditional expectation lecture)

$$\boxed{p_{Y|X}(y \mid x) = \frac{p_{X,Y}(x,y)}{p_X(x)}, \qquad y \in S_x.}$$

It is zero outside that slice. Its normalization follows directly:

$$\sum_{y \in S_x} p_{Y|X}(y \mid x) = \frac{\sum_{y \in S_x} p_{X,Y}(x,y)}{p_X(x)} = 1.$$

The denominator depends on x, but is a constant while summing over y. If $p_X(x) = 0$, the ratio is not defined; a full formal conditional distribution may assign an arbitrary probability law at such an x, but the joint law does not determine it.

Continuous PDFs: the same workflow, with integrals

Assume the pair (X, Y) has a joint PDF, not merely that each marginal has a PDF. Then for $0 < f_X(x) < \infty$, the usual density version is (CMU probability notes)

$$f_X(x) = \int_{S_x} f_{X,Y}(x,y)\, dy,$$

$$\boxed{f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)}, \qquad y \in S_x.}$$

As in the discrete calculation,

$$\int_{S_x} f_{Y|X}(y \mid x)\, dy = \frac{1}{f_X(x)} \int_{S_x} f_{X,Y}(x,y)\, dy = 1.$$

The restriction $0 < f_X(x) < \infty$ makes the displayed ratio meaningful. A density-derived conditional law is determined only for P_X-almost every x, and values on exceptional null sets can be completed separately.

The conditional density describes Y, not X. Thus, for fixed admissible x,

$$P(a < Y \le b \mid X = x) = \int_{(a,b] \cap S_x} f_{Y|X}(y \mid x)\, dy.$$

Why the density ratio is not a zero-over-zero trick

The rigorous justification is the disintegration identity developed below, rather than dividing by $P(X = x)$. An intuitive limiting calculation also helps, provided the required regularity is stated.

For $\varepsilon > 0$ with positive denominator, ordinary conditioning gives

[
P(Y\in B\mid x<X<x+\varepsilon)

\frac{
\int_x^{x+\varepsilon}\int_B f_{X,Y}(u,y),dy,du
}{
\int_x^{x+\varepsilon}f_X(u),du
}.
]

If $u \mapsto \int_B f_{X,Y}(u,y)\,dy$ and $f_X(u)$ are continuous at x, with $f_X(x) > 0$, divide numerator and denominator by ε and let $\varepsilon \downarrow 0$:

$$P(Y \in B \mid x < X < x+\varepsilon) \longrightarrow \frac{\int_B f_{X,Y}(x,y)\,dy}{f_X(x)}.$$

This derives the slice-and-renormalize formula under those assumptions. It does not imply that every approximation to every probability-zero event produces the same conditional law.

Conditional CDFs, quantiles, and reversal

A conditional distribution can also be represented by its CDF:

$$F_{Y|X}(t \mid x) = P(Y \le t \mid X = x).$$

For a PMF, sum the conditional masses with $y \le t$; for a PDF, integrate the conditional density from $-\infty$ to t. For each fixed admissible x, this is an ordinary CDF, including its values 0 below the support and 1 above it; the regular-conditional-distribution framework guarantees a probability law in the Y coordinate (CMU probability notes).

A conditional q-quantile can be defined by

$$Q_{Y|X}(q \mid x) = \inf\{t : F_{Y|X}(t \mid x) \ge q\}, \qquad 0 < q < 1.$$

The product rule reconstructs the joint distribution, and Bayes' rule reverses the conditioning direction (CMU probability notes):

$$f_{X,Y}(x,y) = f_{Y|X}(y \mid x) f_X(x),$$

[
f_{X\mid Y}(x\mid y)

\frac{f_{Y\mid X}(y\mid x)f_X(x)}
{\int f_{Y\mid X}(y\mid u)f_X(u),du}.
]

The denominator is $f_Y(y)$, not a new object to memorize. Replace densities and integration with PMFs and summation when both variables are discrete; use the appropriate mixture of the two when their types differ.

Formal measure-theoretic framework

This section explains what "given $X = x$" means when point probabilities vanish, and why the standard formulas work. It is conceptually important, but the Radon–Nikodym and kernel proofs can be treated as an advanced second pass until the instructor's expectations are clear.

Information as a sigma-algebra

For a real-valued random variable,

$$\sigma(X) = \{X^{-1}(A) : A \in \mathcal{B}(\mathbb{R})\}$$

is the information revealed by observing X. A $\sigma(X)$-measurable real-valued random variable can be written as a measurable function of X, the principle behind the notation $E[Y \mid X] = h(X)$ (CMU probability notes).

Think of $\sigma(X)$ as the collection of questions answerable using X alone. A conditional expectation based on X cannot distinguish outcomes that are indistinguishable through X.

Conditional expectation given information

Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-sigma-algebra and assume $E|Y| < \infty$. A version of $E[Y \mid \mathcal{G}]$ is an integrable random variable Z satisfying both conditions (CMU probability notes):

1. **Measurability:** Z is $\mathcal{G}$-measurable.
2. **Integral matching:** For every $G \in \mathcal{G}$,
$$int_G Z, dP = int_G Y, dP.$$

The first condition says Z uses only the available information. The second says that, within every event distinguishable using that information, Z reproduces the total expected contribution of Y.

For existence, define the finite signed measure

$$\nu(G) = \int_G Y \, dP, \qquad G \in \mathcal{G}.$$

Since $\nu \ll P|_{\mathcal{G}}$, the Radon–Nikodym theorem supplies the derivative $Z = d\nu/d(P|_{\mathcal{G}})$, which is the conditional expectation; it is unique only almost surely (MIT measure-theoretic probability lecture). For nonnegative Y, conditional expectation also has an extended-valued formulation, but this guide uses finite integrable moments unless stated otherwise.

Conditional probability is a special case:

$$P(A \mid \mathcal{G}) = E[\mathbf{1}_A \mid \mathcal{G}].$$

Thus the formal definition is not limited to PMFs or PDFs. It starts from random variables and information, and only later uses densities when they exist.

Conditioning on a variable versus conditioning on a value

Define

$$E[Y \mid X] := E[Y \mid \sigma(X)].$$

There is a Borel function h with $E[Y \mid X] = h(X)$ a.s.; writing $E[Y \mid X = x] = h(x)$ refers to a chosen version of that function (CMU probability notes). The distinction is:

Notation	Kind of object	Interpretation
$E[Y]$	Number	Mean before observing X
$h(x) = E[Y \mid X = x]$	Number at fixed x	Mean under the chosen conditional law at x
h	Deterministic function	Maps observations to conditional means
$h(X) = E[Y \mid X]$	Random variable	Conditional mean before the value of X is revealed

If X is absolutely continuous, changing h at a single value of x does not change $h(X)$ almost surely. A smooth density formula may supply a natural version, but probability theory alone generally does not uniquely specify every isolated point value.

Regular conditional distributions as probability kernels

For real-valued X, Y, a regular conditional distribution of Y given X is a function

$$K : \mathbb{R} \times \mathcal{B}(\mathbb{R}) \to [0, 1]$$

with the following properties (CMU regular conditional probability lecture; CMU probability notes):

1. **Probability law in the second argument:** For every x, $B \mapsto K(x, B)$ is a probability measure.
2. **Measurability in the first argument:** For every Borel B, $x \mapsto K(x, B)$ is measurable.
3. **Disintegration:** For all Borel A, B,
$$boxed P(X in A, Y in B) = int_A K(x, B), P_X(dx).$$

Write $K(x, B) = P(Y \in B \mid X = x)$. The disintegration identity says that averaging the conditional laws over the distribution of X reconstructs the joint law.

For real-valued variables, and more generally standard Borel state spaces, regular conditional distributions exist; arbitrary measurable spaces need additional care (CMU regular conditional probability lecture). Two such kernels are the same probability measure for P_X-almost every x, though their values may differ on a P_X-null set.

Given a regular conditional distribution and $E|Y| < \infty$,

$$h(x) = \int_{\mathbb{R}} y \, K(x, dy)$$

is finite for P_X-almost every x, and $h(X)$ is a version of $E[Y \mid X]$ (CMU probability notes). This one expression covers sums, density integrals, and point masses.

PMFs and PDFs as one Radon–Nikodym construction

Suppose

$$P_{X,Y} \ll \mu \otimes \nu$$

for sigma-finite reference measures μ, ν, and write

$$q(x,y) = \frac{dP_{X,Y}}{d(\mu \otimes \nu)}(x,y).$$

The usual density calculations extend to such product reference measures (CMU probability notes):

- **Discrete pair:** Take counting measure on each variable's finite or countable value set. Then q is a joint PMF.
- **Absolutely continuous pair:** Take Lebesgue measure on each real coordinate. Then q is a joint PDF.
- **Mixed pair:** Take counting measure on the discrete coordinate and Lebesgue measure on the continuous coordinate. Then q is a mixed mass-density.

Tonelli gives the marginal density relative to μ:

$$q_X(x) = \int q(x,y)\, \nu(dy).$$

For $0 < q_X(x) < \infty$, define

$$K(x,B) = \int_B \frac{q(x,y)}{q_X(x)}\, \nu(dy).$$

On the exceptional set, assign a fixed probability law to complete the kernel. To verify the construction, calculate

$$\begin{aligned}
\int_A K(x,B)\, P_X(dx) &= \int_A \left[\int_B \frac{q(x,y)}{q_X(x)}\, \nu(dy) \right] q_X(x)\, \mu(dx) \\
&= \int_A \int_B q(x,y)\, \nu(dy)\, \mu(dx) \\
&= P(X \in A, Y \in B),
\end{aligned}$$

where exceptional sets have zero P_X-mass. The familiar “joint divided by marginal” rule is therefore a consequence of the formal definition.

When densities are not available

Having continuous marginal distributions does not guarantee a joint PDF. For example, if X is uniform on $(-1,1)$ and $Y = X^2$, the joint law is concentrated on a curve, as worked out in the last problem.

The conditional law in that example is still meaningful:

$$K(x, \cdot) = \delta_{x^2},$$

where δ_a is unit point mass at a. A probability kernel is more general than an ordinary density.

Similarly, merely naming a null event does not specify a unique conditional law on it. A sound question specifies the conditioning random variable or information and, if pointwise values or limits matter, the version or regularity convention being used.

A reliable joint-to-conditional derivation procedure

Use the following sequence on every problem. The formulas are short; most mistakes come from skipping a support check or forgetting which variable is fixed.

1. **Identify the types and target.** Are X, Y discrete, continuous with a joint PDF, mixed, or singular? Write the requested object, such as $f_{Y|X}$, before calculating.
2. **Write the entire joint law.** Include its support, indicator, and any unknown normalizing constant.
3. **Verify normalization.** Check nonnegativity and total mass 1. If given $c\,g(x, y)$, solve $c \iint_S g = 1$, or its summation analogue.
4. **Find the relevant slice.** For $Y \mid X = x$, solve the support restrictions for y with x fixed.
5. **Marginalize the conditioning variable.** Compute $p_X(x) = \sum_y p(x, y)$ or $f_X(x) = \int f(x, y)\, dy$.
6. **Divide by that marginal.** Use a positive finite denominator, retain the conditional support, and state zero density or mass outside it.
7. **Verify the conditional law.** Sum or integrate over the unconditioned variable to obtain 1.
8. **Compute the requested quantity.** Use a conditional sum, integral, CDF, or recognized distribution.
9. **Cross-check.** Reconstruct the joint, apply total expectation or total variance, and check whether the answer lies within the possible range.

For an exam solution, use this compact written template:

$$\boxed{\begin{aligned} &S = \{\text{joint restrictions}\}, \qquad S_x = \{\text{allowed values of } y\}, \\ &f_X(x) = \int_{S_x} f_{X,Y}(x, y)\, dy, \\ &f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)} \mathbf{1}_{S_x}(y), \\ &h(x) = E[Y \mid X = x] = \int_{S_x} y\, f_{Y|X}(y \mid x)\, dy. \end{aligned}}$$

This template presumes a joint PDF and admissible x. For discrete variables replace the integrals with sums; for singular laws start with a kernel rather than an unavailable density.

Conditional expectation and the main Part II identities

Calculate a conditional expectation as an ordinary expectation

Once the conditional distribution is known, use its masses or density as the weights (MIT conditional expectation lecture):

$$\boxed{E[Y \mid X = x] = \sum_y y\, p_{Y|X}(y \mid x)}$$

or

$$\boxed{E[Y \mid X = x] = \int_{\mathbb{R}} y\, f_{Y|X}(y \mid x)\, dy.}$$

For $E[g(Y) \mid X = x]$, replace y by $g(y)$, not by a new density. More generally,

$$E[g(X, Y) \mid X = x] = \int g(x, y)\, K(x, dy),$$

under the appropriate integrability assumption. The observed x is held fixed throughout.

For a Bernoulli or indicator-valued Y, the conditional mean is a conditional probability:

$$E[Y \mid X = x] = P(Y = 1 \mid X = x).$$

For a general Y, a mean is only one summary of its distribution. It does not describe the conditional spread or shape.

Total probability and total expectation

A marginal law is a mixture of conditional laws (MIT conditional expectation lecture):

$$P(Y \in B) = \int K(x, B)\, P_X(dx).$$

In particular,

$$p_Y(y) = \sum_x p_{Y|X}(y \mid x) p_X(x),$$

$$f_Y(y) = \int f_{Y|X}(y \mid x) f_X(x)\, dx.$$

For $E|Y| < \infty$, the law of total expectation is (MIT conditional expectation lecture)

$$\boxed{E[E[Y \mid X]] = E[Y].}$$

The continuous derivation explicitly displays the cancellation:

$$\begin{aligned} E[h(X)] &= \int \left[\int y \frac{f_{X,Y}(x,y)}{f_X(x)}\, dy \right] f_X(x)\, dx \\ &= \iint y f_{X,Y}(x,y)\, dy\, dx \\ &= E[Y]. \end{aligned}$$

The measure-theoretic proof is even shorter: take $G = \Omega$ in the defining integral identity. The outer expectation must weight $h(x)$ by the distribution of X; averaging conditional means equally is generally wrong.

Conditional variance and total variance

If $E[Y^2] < \infty$, define (MIT conditional expectation lecture)

$$v(x) = \operatorname{Var}(Y \mid X = x) = E[Y^2 \mid X = x] - h(x)^2.$$

The law of total variance is

$$\boxed{\operatorname{Var}(Y) = E[v(X)] + \operatorname{Var}(h(X)).}$$

To derive it, start with

$$E[Y^2 \mid X] = \operatorname{Var}(Y \mid X) + h(X)^2.$$

Take expectations and subtract $E[Y]^2 = E[h(X)]^2$:

$$\begin{aligned} \operatorname{Var}(Y) &= E[\operatorname{Var}(Y \mid X)] + E[h(X)^2] - E[h(X)]^2 \\ &= E[\operatorname{Var}(Y \mid X)] + \operatorname{Var}(E[Y \mid X]). \end{aligned}$$

The first term is average variation remaining after X is known. The second is variation among the conditional means.

Consequently, $E[\operatorname{Var}(Y \mid X)] \le \operatorname{Var}(Y)$, but this is an average statement. It does not imply $\operatorname{Var}(Y \mid X = x) \le \operatorname{Var}(Y)$ for every individual x.

Algebraic properties, with their assumptions

The following properties hold almost surely, assuming integrability of the displayed variables; the variance and squared-error statements require finite second moments (CMU probability notes).

- **Linearity:** $E[aY + bZ \mid X] = aE[Y \mid X] + bE[Z \mid X]$ for constants a, b.
- **Known quantities remain known:** $E[g(X) \mid X] = g(X)$ if $g(X)$ is integrable.
- **Taking out what is known:** $E[g(X)Y \mid X] = g(X)E[Y \mid X]$, for example when g is bounded and Y is integrable.
- **Nested information:** If $\mathcal{H} \subseteq \mathcal{G}$, then $E[E[Y \mid \mathcal{G}] \mid \mathcal{H}] = E[Y \mid \mathcal{H}]$.
- **More information after less information:** With the same inclusion, $E[E[Y \mid \mathcal{H}] \mid \mathcal{G}] = E[Y \mid \mathcal{H}]$, because the inner result is already known.

- **Independence:** If X and Y are independent, then $E[Y \mid X] = E[Y]$. The converse is false.
- **Conditional Jensen:** For convex ϕ, $\phi(E[Y \mid X]) \le E[\phi(Y) \mid X]$, assuming the required integrability.

Do not apply the tower property to arbitrary nonnested conditioning variables. In general, $E[E[Y \mid X] \mid Z]$ need not equal $E[Y \mid Z]$.

Conditional expectation as the best squared-error predictor

For $Y \in L^2$, let $h(X) = E[Y \mid X]$, and let $g(X)$ be any square-integrable predictor. Conditional expectation minimizes mean squared error among such predictors (MIT conditional expectation lecture).

The proof explains the result. Write

$$Y - g(X) = [Y - h(X)] + [h(X) - g(X)].$$

The cross term has expectation zero because

$$E[(Y - h(X))(h(X) - g(X))] = E[(h(X) - g(X))E[Y - h(X) \mid X]] = 0.$$

Therefore

$$\boxed{E[(Y - g(X))^2] = E[(Y - h(X))^2] + E[(h(X) - g(X))^2].}$$

The second term is nonnegative, and is zero exactly when $g(X) = h(X)$ a.s. This identifies the regression function as $h(x)$; it need not be linear, and a linear regression model may be an approximation rather than the exact conditional mean.

Six fully worked problems

The six problems below are original instructional calculations, with their numerical parameters chosen for this guide. Each includes a derivation and an internal check; the last two also connect the calculations to normal regression and the formal kernel definition.

Discrete joint table: probabilities, conditional moments, and reversal

Problem 1 of 6. Let $X \in \{0, 1\}$ and $Y \in \{0, 1, 2\}$ have the following joint PMF. Find both marginals, both conditional PMFs, $E[Y \mid X = x]$, conditional variances, $P(X = 1 \mid Y \ge 1)$, and verify total expectation and total variance.

$y \backslash x$	$x = 0$	$x = 1$
$y = 0$	$2/20$	$1/20$
$y = 1$	$3/20$	$4/20$
$y = 2$	$1/20$	$9/20$

1. **Validate the joint PMF.** Every entry is nonnegative, and the numerator total is $2 + 1 + 3 + 4 + 1 + 9 = 20$. Thus the table has total probability 1.

2. **Calculate marginals.** Sum down columns to retain X, and across rows to retain Y:

$$p_X(0) = \frac{6}{20} = \frac{3}{10}, \qquad p_X(1) = \frac{14}{20} = \frac{7}{10},$$
$$p_Y(0) = \frac{3}{20}, \qquad p_Y(1) = \frac{7}{20}, \qquad p_Y(2) = \frac{10}{20} = \frac{1}{2}.$$

3. **Find $Y \mid X = x$.** Divide each column by its marginal:

y	$p_{Y\mid X}(y \mid 0)$	$p_{Y\mid X}(y \mid 1)$
0	$2/6 = 1/3$	$1/14$
1	$3/6 = 1/2$	$4/14 = 2/7$
2	$1/6$	$9/14$

   The first column sums to $1/3 + 1/2 + 1/6 = 1$. The second sums to $(1 + 4 + 9)/14 = 1$.

4. **Compute conditional means.**

$$h(0) = 0\left(\frac{1}{3}\right) + 1\left(\frac{1}{2}\right) + 2\left(\frac{1}{6}\right) = \frac{5}{6},$$
$$h(1) = 0\left(\frac{1}{14}\right) + 1\left(\frac{4}{14}\right) + 2\left(\frac{9}{14}\right) = \frac{11}{7}.$$

   Since X takes only 0 and 1, a convenient expression is

$$E[Y mid X] = frac56 + frac3142X.$$

   This random variable takes two possible values; neither value is itself the marginal mean of Y.

5. **Compute conditional second moments and variances.**

$$E[Y^2 \mid X = 0] = \frac{1}{2} + 4\left(\frac{1}{6}\right) = \frac{7}{6},$$
$$v(0) = \frac{7}{6} - \left(\frac{5}{6}\right)^2 = \frac{42 - 25}{36} = \frac{17}{36}.$$
$$E[Y^2 \mid X = 1] = \frac{4}{14} + 4\left(\frac{9}{14}\right) = \frac{20}{7},$$
$$v(1) = \frac{20}{7} - \left(\frac{11}{7}\right)^2 = \frac{140 - 121}{49} = \frac{19}{49}.$$

6. **Reverse the conditioning.** Divide within each row, not within each column:

y	$p_{X\mid Y}(0 \mid y)$	$p_{X\mid Y}(1 \mid y) = E[X \mid Y = y]$
0	$2/3$	$1/3$
1	$3/7$	$4/7$
2	$1/10$	$9/10$

   For the event $Y \geq 1$, add the relevant joint masses before dividing:

$$P(X = 1 mid Y ge 1) = frac(4 + 9)/20(3 + 4 + 1 + 9)/20 = frac1317.$$

7. **Check total expectation.**

$$E[h(X)] = \frac{3}{10}\frac{5}{6} + \frac{7}{10}\frac{11}{7} = \frac{1}{4} + \frac{11}{10} = \frac{27}{20}.$$

   Directly,

$$E[Y] = 0frac320 + 1frac720 + 2frac12 = frac2720.$$

8. **Check total variance.** Direct calculation gives
$$E[Y^2] = frac720 + 4frac12 = frac4720,$$
$$operatornameVar(Y) = frac4720 - left(frac2720right)^2 = frac211400.$$
The conditional decomposition gives
$$E[v(X)] = frac310frac1736 + frac710frac1949 = frac347840,$$
$$operatornameVar(h(X)) = left(frac3142right)^2 frac310frac710 = frac9618400.$$
Their sum is
$$frac3470 + 9618400 = frac44318400 = frac211400.$$

The variables are dependent: for instance, $P(Y = 2 \mid X = 1) = 9/14 \neq 1/2 = P(Y = 2)$. The essential operation is renormalization in the direction selected by the conditioning statement.

Polynomial density on a square: conditional CDF and nonlinear mean

Problem 2 of 6. Suppose

$$f_{X,Y}(x, y) = c(x^2 + y^2), \qquad 0 < x < 1, \quad 0 < y < 1,$$

and zero otherwise. Find c, both marginals, $f_{Y|X}$, its CDF and conditional mean, and evaluate $P(Y \le 1/2 \mid X = 1/2)$ and $\operatorname{Var}(Y \mid X = 1/2)$.

1. **Normalize the joint density.**
$$1 = c \int_0^1 \int_0^1 (x^2 + y^2)\, dy\, dx = c\left(\frac{1}{3} + \frac{1}{3}\right).$$
Hence $c = 3/2$. The density is nonnegative on its rectangular support.

2. **Calculate the marginals.**
$$f_X(x) = \frac{3}{2}\int_0^1 (x^2 + y^2)\, dy = \frac{3}{2}\left(x^2 + \frac{1}{3}\right) = \frac{3x^2 + 1}{2}, \qquad 0 < x < 1.$$
By symmetry,
Each marginal is zero outside its interval and integrates to 1.

3. **Divide and check the conditional density.**
$$f_{Y|X}(y \mid x) = \frac{x^2 + y^2}{x^2 + 1/3}, \qquad 0 < y < 1.$$
For each $0 < x < 1$,
$$int_0^1 fracx^2 + y^2x^2 + 1/3, dy = fracx^2 + 1/3x^2 + 1/3 = 1.$$

4. **Build the full conditional CDF.**
$$F_{Y|X}(t \mid x) = \begin{cases} 0, & t \le 0, \\ \dfrac{x^2 t + t^3/3}{x^2 + 1/3}, & 0 < t < 1, \\ 1, & t \ge 1. \end{cases}$$
At $x = 1/2$, $t = 1/2$,
$$P(Yle1/2midX = 1/2) = frac1/8 + 1/241/4 + 1/3 = frac1/67/12 = frac27.$$

5. **Calculate the conditional mean.**
$$h(x) = \frac{1}{x^2 + 1/3}\int_0^1 y(x^2 + y^2)\, dy = \frac{x^2/2 + 1/4}{x^2 + 1/3}.$$
At $x = 1/2$,

$$E[Y mid X = 1/2] = frac1/8 + 1/47/12 = frac3/87/12 = frac914.$$

The conditional mean is a rational function, not a linear function of x.

6. **Calculate the conditional variance.**

$$E[Y^2 \mid X = x] = \frac{x^2/3 + 1/5}{x^2 + 1/3}.$$

At $x = 1/2$, this becomes

$$frac1/12 + 1/57/12 = frac1735.$$

Therefore

$operatornameVar(Y mid X = 1/2) = frac1735 - left(frac914right)^2 = frac476 - 405980$

7. **Verify total expectation.**

$$\begin{aligned} E[h(X)] &= \int_0^1 \frac{x^2/2 + 1/4}{x^2 + 1/3}\,\frac{3}{2}(x^2 + 1/3)\,dx \\ &= \frac{3}{2}\left(\frac{1}{6} + \frac{1}{4}\right) = \frac{5}{8}. \end{aligned}$$

The marginal calculation agrees:

$$E[Y] = int_0^1 y frac3y^2 + 12, dy = frac12 left(frac34 + frac12right) = frac58.$$

The support is rectangular, yet the variables are dependent because the conditional density changes with x. As x increases, the constant-in-y contribution becomes more prominent, changing the shape of the conditional distribution.

Piecewise support: geometry, total variance, and mean independence

Problem 3 of 6. Let $f_{X,Y}(x, y) = 1$ on

$$S = \{(x, y) : 0 < x < 2,\ 0 < y < \min(x, 2 - x)\},$$

and zero elsewhere. Find both marginals, both conditional distributions, $E[Y \mid X = x]$, the conditional CDF of Y, and the total-variance decomposition of Y.

1. **Understand the region and normalize.** The support is the triangle with vertices $(0, 0), (1, 1), (2, 0)$, apart from its boundary. Write its vertical height as
   Its area is

$$int_0^1 x, dx + int_1^2 (2 - x), dx = frac12 + frac12 = 1,$$

   so the constant density 1 is valid.

2. **Calculate both marginals from their slices.**

$$f_X(x) = \int_0^{m(x)} 1\,dy = m(x), \qquad 0 < x < 2.$$

   For fixed y, the inequalities are $y < x < 2 - y$, with $0 < y < 1$. Therefore

3. **Find and validate both conditional densities.**

$$\begin{aligned} f_{Y|X}(y \mid x) &= \frac{1}{m(x)}, \qquad 0 < y < m(x), \\ f_{X|Y}(x \mid y) &= \frac{1}{2(1 - y)}, \qquad y < x < 2 - y. \end{aligned}$$

   Both integrate to 1 because density height times interval length equals 1. Thus
   $Y mid X = x simoperatornameUniform(0, m(x)), qquad X mid Y = y simoperatornameUni$

4. **Derive the conditional mean and variance of Y.**

$$h(x) = \int_0^{m(x)} \frac{y}{m(x)}\, dy = \frac{m(x)}{2},$$
$$E[Y^2 \mid X = x] = \int_0^{m(x)} \frac{y^2}{m(x)}\, dy = \frac{m(x)^2}{3}.$$

Subtracting the square of the mean gives

$$v(x) = fracm(x)^23 - fracm(x)^24 = fracm(x)^212.$$

5. **Write the CDF and evaluate a probability.**

$$F_{Y|X}(t \mid x) = \begin{cases} 0, & t \le 0, \\ t/m(x), & 0 < t < m(x), \\ 1, & t \ge m(x). \end{cases}$$

At $x = 3/2$, $m(x) = 1/2$, so

$$E[YmidX = 3/2] = frac14,$$
$$P(Y > 1/4midX = 3/2) = int_1/4^1/2 2, dy = frac12.$$

Notice that the conditional density equals 2, while the probability remains between 0 and 1.

6. **Check total expectation.** Symmetry around $x = 1$ simplifies the piecewise integral:

$$E[h(X)] = 2int_0^1 fracx2x, dx = int_0^1 x^2, dx = frac13.$$

Directly,

$$E[Y] = 2int_0^1 y(1-y), dy = 2left(frac12 - frac13right) = frac13.$$

7. **Check total variance.** From the marginal,

$$E[Y^2] = 2int_0^1 (y^2 - y^3), dy = 2left(frac13 - frac14right) = frac16,$$
$$operatornameVar(Y) = frac16 - frac19 = frac118.$$

The within-conditional component is

$$E[v(X)] = 2int_0^1 fracx^212x, dx = frac16cdotfrac14 = frac124.$$

The between-means component is

$$E[h(X)^2] = 2int_0^1 fracx^24x, dx = frac18,$$
$$operatornameVar(h(X)) = frac18 - left(frac13right)^2 = frac172.$$

Together,

$$frac124 + frac172 = frac118.$$

8. **Inspect the reverse conditional mean.** Although the interval for $X \mid Y = y$ changes with y, its midpoint does not:

$$E[XmidY = y] = fracy + (2-y)2 = 1.$$

Yet

$$operatornameVar(XmidY = y) = frac[2(1-y)]^212 = frac(1-y)^23,$$

which changes with y.

This is a concrete counterexample to "constant conditional mean implies independence." Here $E[X \mid Y] = E[X] = 1$, but the full conditional distribution of X plainly depends on Y.

Mixed discrete–continuous model: Bayes reversal and a beta posterior

Problem 4 of 6. Let a random success probability Y have density $f_Y(y) = 6y(1-y)$ for $0 < y < 1$, and suppose

$$X \mid Y = y \sim \text{Binomial}(4, y).$$

Find the mixed joint law, marginal PMF of X, conditional PDF of $Y \mid X = x$, conditional moments, and $P(Y > 1/2 \mid X = 3)$. This is a beta-binomial construction, whose posterior family is beta (beta-binomial course handout); the calculations below derive the specific result rather than assume it.

1. **Identify the reference measures.** X is discrete on $\{0,1,2,3,4\}$; Y is continuous on $(0,1)$. The mixed joint mass-density is
$$q(x,y) = P(X = x mid Y = y) f_Y(y) = 6 binom{4}{x} y^{x+1}(1-y)^{5-x}.$$
Its meaning is
$$P(X = x, Y in B) = int_B q(x,y), dy,$$
not $P(X = x, Y = y) = q(x,y)$.

2. **Check joint normalization.** The binomial probabilities sum to 1 at every y, and the prior density integrates to 1:
$$sum_{x=0}^{4} int_0^1 q(x,y), dy = int_0^1 6y(1-y) sum_{x=0}^{4} binom{4}{x} y^{x}(1-y)^{4-x}, dy = 1.$$

3. **Integrate out Y to obtain p_X.** Define the beta integral
$$B(a,b) = int_0^1 y^{a-1}(1-y)^{b-1}, dy.$$
For positive integers, $B(a,b) = (a-1)!(b-1)!/(a+b-1)!$. Hence

x	0	1	2	3	4
$p_X(x)$	$5/35$	$8/35$	$9/35$	$8/35$	$5/35$

The numerators sum to 35. This marginal is not generally a binomial PMF with its success probability replaced by $E[Y]$.

4. **Divide to reverse the conditioning.**
$$\begin{aligned} f_{Y \mid X}(y \mid x) &= \frac{q(x,y)}{p_X(x)} \\ &= \frac{y^{x+1}(1-y)^{5-x}}{B(x+2, 6-x)}, \qquad 0 < y < 1. \end{aligned}$$
The beta integral makes the conditional density integrate to 1. Matching its exponents gives
$$boxed{Y mid X = x} sim operatorname{Beta}(x+2, 6-x).$$

5. **Derive conditional moments from integrals.**
$$\begin{aligned} E[Y \mid X = x] &= \frac{B(x+3, 6-x)}{B(x+2, 6-x)} = \frac{x+2}{8}, \\ E[Y^2 \mid X = x] &= \frac{B(x+4, 6-x)}{B(x+2, 6-x)} = \frac{(x+2)(x+3)}{8 \cdot 9}. \end{aligned}$$
Therefore
In particular,
$E[Y mid X = 3] = frac{5}{8}, qquad operatorname{Var}(Y mid X = 3) = frac{5}{cdot 3}{576} = frac{5}$

6. **Calculate a posterior probability explicitly.** At $x = 3$, the density is
   Thus

7. **Check total expectation.** The marginal table gives
   $$E[X] = frac0cdot5 + 1cdot8 + 2cdot9 + 3cdot8 + 4cdot535 = 2.$$
   Consequently
   =E\left[\frac{X+2}{8}\right]
   =\frac{2+2}{8}=\frac12.
   ]
   Directly,
   $$E[Y] = 6int_0^1(y^2 - y^3), dy = 6left(frac13 - frac14right) = frac12.$$

8. **Check total variance.** From the marginal PMF,
   $$E[X^2] = frac8 + 36 + 72 + 8035 = frac285, qquadoperatornameVar(X) = frac85.$$
   Therefore
   $$operatornameVar(E[YmidX]) = fracoperatornameVar(X)64 = frac140.$$
   Since $(X + 2)(6 - X) = -X^2 + 4X + 12$,
   $$E[v(X)] = frac-28/5 + 8 + 12576 = frac140.$$
   Direct calculation from f_Y gives
   $E[Y^2] = 6left(frac14 - frac15right) = frac310, quadoperatornameVar(Y) = frac310 -$
   agreeing with $1/40 + 1/40$.

The conditional mean can also be written as

$$fracx + 28 = frac12left(frac12right) + frac12left(fracx4right).$$

For this particular model it averages the prior mean and observed success fraction equally. Most importantly, Bayes reversal used a PMF in the denominator and a mixed joint mass-density in the numerator.

Bivariate normal: deriving the conditional law by completing the square

Problem 5 of 6. Suppose (X, Y) is jointly normal, with positive standard deviations σ_X, σ_Y and $|\rho| < 1$. Derive $Y \mid X = x$ from the joint PDF, then evaluate the result for $\mu_X = 10, \mu_Y = 50, \sigma_X = 2, \sigma_Y = 6, \rho = 1/2$.

The normal conditional mean and variance are standard results (Penn State bivariate normal lesson). The purpose here is to recover them from the density rather than memorize them.

1. **Write the joint density.** Define standardized coordinates
   $$u = fracx - mu_Xsigma_X, qquadv = fracy - mu_Ysigma_Y.$$
   The joint PDF is
   $f_X, Y(x, y) = frac12pisigma_Xsigma_Ysqrt1 - rho^2expleft[-fracu^2 - 2rhouv + v^22(1 -$
   with support all of $\mathbb{R}^2$.

2. **Complete the square in the variable not conditioned on.** With x, and thus u, fixed,
   $$u^2 - 2rhouv + v^2 = (v - rhou)^2 + (1 - rho^2)u^2.$$
   Hence

3. **Obtain the marginal by integration.** Set

$$z = frac{y - mu_Y - rho(sigma_Y/sigma_X)(x - mu_X)}{sigma_Y sqrt{1 - rho^2}}.$$

Here $\phi(z) = (2\pi)^{-1/2} e^{-z^2/2}$ is the standard normal PDF, and $\Phi(t) = \int_{-\infty}^{t} \phi(z)\, dz$ is its CDF. The second factor integrates to $\int_{\mathbb{R}} \phi(z)\, dz = 1$, so

$$f_X(x) = frac{1}{sqrt{2pi}sigma_X} exp left[-frac{(x - mu_X)^2}{2sigma_X^2} right].$$

It integrates to 1 as well, so this factorization also verifies joint normalization.

4. # Divide by f_X. The first factor cancels, leaving [f_{Y\mid X}(y\mid x)

\frac{1}{\sqrt{2\pi}\sigma_Y\sqrt{1-\rho^2}}
\exp\left

$$-frac{y - mu_Y - rho(sigma_Y/sigma_X)(x - mu_X)^2}{2sigma_Y^2(1 - rho^2)} right$$

.
]
Thus, using $N(\text{mean, variance})$ notation,
$$boxed{Y mid X = x sim N left(mu_Y + rho frac{sigma_Y}{sigma_X}(x - mu_X), sigma_Y^2(1 - rho^2)}$$

5. **Compute the numerical conditional moments.**

$$E[Y mid X = x] = 50 + frac{12}{62}(x - 10) = 50 + 1.5(x - 10),$$
$$operatorname{Var}(Y mid X = x) = 36 left(1 - frac{14}{right) = 27.}$$

At $x = 12$,

$$Y mid X = 12 sim N(53, 27),$$

with conditional standard deviation $\sqrt{27} = 3\sqrt{3}$, not 27.

6. **Calculate a conditional probability.**

7. **Check both total-moment identities.**
=50+1.5(E[X]-10)=50,
]

$$E[operatorname{Var}(Y mid X)] = 27,$$
$$operatorname{Var}(E[Y mid X]) = 1.5^2 operatorname{Var}(X) = 2.25 cdot 4 = 9.$$

Therefore

$$operatorname{Var}(Y) = 27 + 9 = 36 = sigma_Y^2.$$

Joint normality is essential: normal marginal distributions alone do not establish this conditional formula. Within the jointly normal family, $\rho = 0$ gives independence; at $|\rho| = 1$ the law is degenerate, so the two-dimensional PDF used above no longer applies (Penn State multivariate conditional distributions; Penn State bivariate normal lesson).

Singular joint law: deterministic conditioning and a probability kernel

Problem 6 of 6. Let $X \sim \text{Uniform}(-1, 1)$ and $Y = X^2$. Find the marginal distribution of Y, explain why there is no joint Lebesgue PDF, construct $Y \mid X = x$, find $X \mid Y = y$, and verify the formal definitions and moment identities.

1. **Find the marginal CDF of Y.** Since $0 \le X^2 \le 1$,
The middle probability is interval length $2\sqrt{t}$ divided by the total length 2.

2. **Differentiate to obtain the marginal PDF.**
   It is unbounded near zero but integrable:
   $$int_0^1 frac12sqrty, dy = [sqrty]_0^1 = 1.$$
   Thus both X and Y have marginal PDFs.

3. **Explain why there is no ordinary joint PDF.** The curve
   has two-dimensional Lebesgue measure zero, but $P((X,Y)\in C)=1$. If a joint Lebesgue PDF existed, its integral over C would be zero, a contradiction.

4. **Construct the conditional law of Y.** Once $X=x$ is known, $Y=x^2$ is known exactly.
   Choose
   $$K(x,B) = mathbf1_B(x^2), qquad K(x,cdot) = delta_{x^2}.$$
   The conditional CDF and moments are
   $$F_{YmidX}(tmidx) = mathbf1_{tgex^2},$$
   $$E[YmidX=x] = x^2, qquad operatorname{Var}(YmidX=x) = 0.$$
   This conditional law has a point mass, not an ordinary Lebesgue PDF.

5. **Verify disintegration directly.** For Borel sets A, B,
   Measurability follows because $x\mapsto x^2$ is continuous, and δ_{x^2} is a probability measure at each x. All kernel requirements are satisfied.

6. **Verify the formal conditional expectation.** Set $Z=X^2$. It is $\sigma(X)$-measurable, integrable, and $Z=Y$, so for every $G\in\sigma(X)$,
   $$int_G Z, dP = int_G Y, dP.$$
   Therefore $E[Y\mid X]=X^2$ follows from the formal definition, without any density ratio.

7. **Reverse the conditioning.** For $0<y<1$, the possible X-values are $-\sqrt{y}$ and $+\sqrt{y}$. Each inverse branch contributes marginal density
   $$frac12left|fracd(pmsqrty)dyright| = frac14sqrty,$$
   so dividing each contribution by $f_Y(y)=1/(2\sqrt{y})$ gives probability $1/2$. Thus
   $$XmidY = ysimfrac12delta_{-sqrty} + frac12delta_{sqrty}.$$
   Consequently
   $E[XmidY=y]=0, qquad E[X^2midY=y]=y, qquad operatorname{Var}(XmidY=y)=y$

8. **Check total expectation and total variance.**
   $$E[Y] = E[X^2] = frac12int_{-1}^1 x^2, dx = frac13,$$
   $$E[Y^2] = E[X^4] = frac12int_{-1}^1 x^4, dx = frac15,$$
   $$operatorname{Var}(Y) = frac15 - frac19 = frac445.$$
   Since Y is determined by X,
   $E[operatorname{Var}(YmidX)] + operatorname{Var}(E[YmidX]) = 0 + operatorname{Var}$
   In the reverse direction,
   $operatorname{Var}(X) = E[operatorname{Var}(XmidY)] + operatorname{Var}(E[XmidY])$

9. **See the meaning of a version.** The natural kernel gives $K(0,\cdot)=\delta_0$. Replace this one value with $\widetilde K(0,\cdot)=\delta_1$, leaving the kernel unchanged elsewhere.
   Since $P_X(\{0\})=0$, this altered kernel still satisfies disintegration. Its conditional mean function differs at $x=0$, but the two corresponding random variables agree almost surely.

The natural deterministic version is the useful one here, but the last step shows why pointwise uniqueness must not be claimed without an additional convention. This example also proves

that zero covariance and a constant reverse conditional mean do not imply independence: $E[X \mid Y] = 0$, yet Y is a nonconstant function of X.

Lecture preparation and self-testing

A practical study sequence

Use three passes rather than trying to memorize the entire document at once. In each pass, explain the denominator and the support aloud before doing algebra.

- **Foundations pass:** Review PMF versus PDF, joint versus marginal, slices, and $E[Y \mid X = x]$ versus $E[Y \mid X]$. Rework the discrete table and square-density problems.
- **Part II calculation pass:** Derive the conditional CDF, total expectation, and total variance. Rework the piecewise-support and mixed-model problems without looking at the bounds.
- **Formal and synthesis pass:** State the two conditions defining conditional expectation and the three kernel conditions. Rework the normal square completion and singular-law example.

If time is limited before class, prioritize the derivation procedure, the total-moment identities, and problems 1–3. Use problems 4–6 to prepare for Bayes, regression, and measure-theoretic extensions rather than assume all will appear in the next lecture.

Questions worth bringing to lecture

- **Course scope:** Which formal definitions and existence results are examinable, and which are enrichment?
- **Conditioning convention:** Are pointwise conditional densities treated through their natural continuous versions, with almost-everywhere qualifications understood?
- **Support handling:** Should piecewise supports be sketched, expressed with indicators, or written as integration bounds?
- **Conditional moments:** Will problems require total variance and regression interpretation, or primarily direct conditional integrals?
- **Model types:** Are mixed and singular examples included, or is the focus on joint PMFs and joint PDFs?
- **Normal models:** Is the bivariate-normal formula to be memorized, derived by square completion, or both?

Common errors and their repairs

Error	Repair
Divide $f_{Y\|X}$ by f_Y	Divide by the marginal of the variable after the conditioning bar: f_X.
Integrate over x to obtain $f_X(x)$	Integrate out y, retaining x.
Use marginal bounds for a conditional integral	Recompute the slice S_x; its bounds may depend on x.

Error	Repair
Drop a support indicator after algebraic factorization	Factor the full density, including where it is zero.
Treat $f_{Y\|X}(y \mid x)$ as $P(Y = y \mid X = x)$	Integrate a conditional PDF over an event to obtain probability.
Treat $E[Y \mid X]$ as a fixed number	Write $h(X)$, then distinguish it from $h(x)$.
Compute $E[Y^2 \mid X = x]$ and call it variance	Subtract $E[Y \mid X = x]^2$.
Average conditional means without marginal weights	Use $\sum_x h(x)p_X(x)$ or $\int h(x)f_X(x)\,dx$.
Conclude independence from a constant conditional mean	Check the entire conditional law or the joint factorization.
Assume normal marginals imply joint normality	Require the joint normal model explicitly.
Apply the tower property without nested information	Check the relevant sigma-algebra inclusion first.
Assume a joint PDF exists because both marginal PDFs exist	Check whether the joint law is singular or otherwise lacks a density.
Set an undefined conditional ratio to zero everywhere	A conditional law must still have total mass 1; use a proper arbitrary law on exceptional null sets if a complete kernel is needed.
Claim conditioning always lowers variance at every x	Total variance gives the reduction only after averaging over X.

Retrieval questions

Try answering without looking back, then compare against the answer key. These are short conceptual checks, separate from the six fully worked problems.

1. Why must $p_X(x)$, rather than $p_Y(y)$, appear in $p_{Y|X}(y \mid x)$?
2. For support $0 < y < x < 1$, what are the integration bounds for $f_X(x)$ and $f_Y(y)$?
3. Is $F_{Y|X}(t \mid x) = F_{X,Y}(x, t)/F_X(x)$ the correct general formula?
4. Why can a conditional PDF exceed 1?
5. What is the difference among h, $h(x)$, and $h(X)$?
6. Which event in the definition of conditional expectation proves $E[E[Y \mid \mathcal{G}]] = E[Y]$?
7. What extra moment condition is needed for total variance and the best squared-error predictor result?
8. Does $E[Y \mid X] = E[Y]$ imply independence?
9. Can both marginals have PDFs while their joint law has no two-dimensional PDF?
10. What does the disintegration identity reconstruct?
11. Why does changing a conditional law at one continuous x-value not necessarily change the joint law?

12. Which variable remains random inside the calculation $E[Y \mid X = x]$?

Short answer key

1. **Normalization:** The denominator is the total mass of the $X = x$ slice, so conditional probabilities over y sum to 1.
2. **Bounds:** $f_X(x) = \int_0^x f(x, y)\, dy$ for $0 < x < 1$, while $f_Y(y) = \int_y^1 f(x, y)\, dx$ for $0 < y < 1$.
3. **Wrong conditioning event:** Where the denominator is positive, that CDF ratio describes $P(Y \le t \mid X \le x)$, not generally $P(Y \le t \mid X = x)$.
4. **Density versus probability:** Density height is not probability; its integral over an event is.
5. **Three objects:** h is a function, $h(x)$ is its value at a fixed argument, and $h(X)$ is a random variable.
6. **Whole space:** Choose $G = \Omega$.
7. **Finite second moment:** Assume $E[Y^2] < \infty$.
8. **No:** It establishes mean independence in that direction, not full independence.
9. **Yes:** The relation $Y = X^2$ in problem 6 provides an example.
10. **Joint law:** Averaging conditional laws over P_X recovers probabilities involving both X and Y.
11. **Null-set freedom:** A singleton with zero P_X-mass contributes nothing to the disintegration integral.
12. **The target variable:** Y varies under its conditional law; x is held fixed.

Compact reference sheet

The formulas below collect the working results derived above. Keep the support, reference measure, and integrability assumptions alongside them rather than treating them as context-free algebra.

Joint, marginal, and conditional

$$p_X(x) = \sum_y p_{X,Y}(x, y), \qquad p_{Y|X}(y \mid x) = \frac{p_{X,Y}(x, y)}{p_X(x)} \quad [p_X(x) > 0].$$

$$f_X(x) = \int f_{X,Y}(x, y)\, dy, \qquad f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)} \quad [0 < f_X(x) < \infty].$$

$$P(X \in A, Y \in B) = \int_A K(x, B)\, P_X(dx).$$

Conditional moments

$$h(x) = E[Y \mid X = x] = \sum_y y\, p_{Y|X}(y \mid x) \quad \text{or} \quad \int y\, f_{Y|X}(y \mid x)\, dy.$$

$$v(x) = E[Y^2 \mid X = x] - h(x)^2.$$

$$E[Y \mid X] = h(X), \qquad E[h(X)] = E[Y].$$

$$\operatorname{Var}(Y) = E[v(X)] + \operatorname{Var}(h(X)).$$

Formal conditional expectation

$$Z = E[Y \mid \mathcal{G}] \iff \begin{cases} Z \text{ is integrable and } \mathcal{G}\text{-measurable,} \\ \int_G Z\, dP = \int_G Y\, dP \quad \text{for every } G \in \mathcal{G}, \end{cases}$$

assuming $Y \in L^1$. Versions are unique a.s.; $h(x)$ and $K(x, \cdot)$ are determined only P_X-almost everywhere.

The final check on every calculation is: **What is fixed, what still varies, and does the resulting distribution have total mass 1?** Once those are correct, the rest is ordinary probability and integration.

Verification note

The six problems' key algebraic results were checked with 54 exact symbolic assertions, including normalizations, selected conditional means and variances, probability integrals, and total-moment identities. The normal-tail value was evaluated numerically as 0.12410653949496186; the hand derivations above remain the primary study material, while computation serves as a cross-check.